



Faculty of Engineering & Technology
Electrical & Computer Engineering Department

ENCS4370 - Computer Architecture
Project 1
Medical Test Management System

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Section: 1

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➔ Test #1:

This test will not contain any errors to be handled

- The content of the file

```
1300524: HGB, 2024-09, 33.0
1300527: HGB, 2024-10, 17.0
1300600: HGB, 2025-08, 104.87
1300525: BGT, 2024-09, 13.52
1300526: BGT, 2024-10, 55.79
1300528: BGT, 2024-11, 25.97
1300529: BGT, 2024-12, 80.0
1300529: LDL, 2023-12, 91.0
1300530: LDL, 2020-06, 150.0
1300531: LDL, 2024-02, 83.50
1500500: BPT, 2024-08, 150.0, 33.79
1300500: BPT, 2024-03, 13.50, 120.0
1300511: BPT, 2024-04, 111.0, 33.50
1300530: BPT, 2022-06, 99.0, 150.0
1300531: BPT, 2024-01, 13.50, 100.0
1300600: BPT, 2024-12, 100.0, 70.0
```

1- Test Option1:

```
Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
1
please enter the patient ID number : (7-digit number)
1600489
please enter the test type : ( BBB any other formate will not be accepted ! )
LDL
please enter the test date : (YYYY-MM any other formate will not be accepted ! )
2021-08
please enter the test result : (x.x any other format will not be accepted)
18.9

Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
1
please enter the patient ID number : (7-digit number)
1600489
please enter the test type : ( BBB any other formate will not be accepted ! )
HGB
please enter the test date : (YYYY-MM any other formate will not be accepted ! )
2021-09
please enter the test result : (x.x any other format will not be accepted)
50.1
```

```

Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
1
please enter the patient ID number : (7-digit number)
1600489
please enter the test type : ( BBB any other formate will not be accepted ! )
BGT
please enter the test date : (YYYY-MM any other formate will not be accepted ! )
2021-10
please enter the test result : (x.x any other format will not be accepted)
80.88

Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
1
please enter the patient ID number : (7-digit number)
1600489
please enter the test type : ( BBB any other formate will not be accepted ! )
BPT
please enter the test date : (YYYY-MM any other formate will not be accepted ! )
2021-11
please enter the test result : (x.x any other format will not be accepted)
121
please enter the diastolic blood pressure : (x.x any other format will not be accepted !!)70

```

2- Test Option2:

```

Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
2
please enter the patient ID number : (7-digit number)
1600489
please enter your selection :
1-Retrieve all patient tests.
2-Retrieve all up normal patient tests
3-Retrieve all patient tests in a given specific period.
1
1600489: HGB, 2021-9, 50.1
1600489: BGT, 2021-10, 80.88
1600489: LDL, 2021-8, 18.9
1600489: BPT, 2021-11, 121.0, 70.0

Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
2
please enter the patient ID number : (7-digit number)
1600489
please enter your selection :
1-Retrieve all patient tests.
2-Retrieve all up normal patient tests
3-Retrieve all patient tests in a given specific period.
2
1600489: HGB, 2021-9, 50.1
1600489: BPT, 2021-11, 121.0, 70.0

```

```

Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
2
please enter the patient ID number : (7-digit number)
1600489
please enter your selection :
1-Retrieve all patient tests.
2-Retrieve all up normal patient tests
3-Retrieve all patient tests in a given specific period.
3
please enter the starting date (YYYY-MM) :
2021-06
please enter the ending date (YYYY-MM) :
2021-09
1600489: HGB, 2021-9, 50.1
1600489: LDL, 2021-8, 18.9

```

3- Test Option3:

```

Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
3
Enter the name of the test: HGB
1600489: Hgb, 2021-09, 50.1
1300524: Hgb, 2024-09, 33.0
1300600: Hgb, 2025-08, 104.87
Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
3
Enter the name of the test: BGT
1300525: BGT, 2024-09, 13.52
1300526: BGT, 2024-10, 55.79
1300528: BGT, 2024-11, 25.96

```

```

Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
3
Enter the name of the test: LDL
1600489: LDL, 2021-08, 18.9
1300529: LDL, 2023-12, 91.0
1300531: LDL, 2024-02, 83.5
Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
3
Enter the name of the test: BPT
1600489: BPT, 2021-11, 121.0, 70.0
1500500: BPT, 2024-08, 150.0, 33.79
1300500: BPT, 2024-03, 13.5, 120.0
1300511: BPT, 2024-04, 111.0, 33.5
1300530: BPT, 2022-06, 99.0, 150.0
1300531: BPT, 2024-01, 13.5, 100.0
1300600: BPT, 2024-12, 100.0, 70.0

```

4- Test Option4:

```

Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
4
The average value of the HGB test is: 51.2425
The average value of the BGT test is: 51.232002
The average value of the LDL test is: 85.85
The average value of the first BPT result (Systolic Blood Pressure) is: 86.85714
The average value of the second BPT result (Diastolic Blood Pressure) is: 103.89857

```

5- Test Option5:

```
Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
5
Enter the ID:1600489
TestName:BGT
Date:2021-10
Please enter the NewResult: 56.4
The result has updated successfully
```

6- Test Option6 & Option7:

```
Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
6
ID:1600489
TestName:BPT
Date:2021-11
The test has been deleted successfully

Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
7
Do you want to save changes?
Enter 1 if yes, 0 if no: 1

-- program is finished running --
```

7- The file after writing on it:

```
1600489: HGB, 2021-09, 50.9
1300600: HGB, 2025-08, 104.87
1300527: HGB, 2024-10, 17.0
1300524: HGB, 2024-09, 33.0
1600489: BGT, 2021-10, 56.40
1300529: BGT, 2024-12, 80.0
1300528: BGT, 2024-11, 25.96
1300526: BGT, 2024-10, 55.79
1300525: BGT, 2024-09, 13.52
1600489: LDL, 2021-08, 18.89
1300531: LDL, 2024-02, 83.50
1300530: LDL, 2020-06, 150.0
1300529: LDL, 2023-12, 91.0
1300600: BPT, 2024-12, 100.0, 70.0
1300531: BPT, 2024-01, 13.50, 100.0
1300530: BPT, 2022-06, 99.0, 150.0
1300511: BPT, 2024-04, 111.0, 33.50
1300500: BPT, 2024-03, 13.50, 120.0
1500500: BPT, 2024-08, 150.0, 33.79|
```

→ Test #2:

This test will contain sum issues and errors in the input and will be handled by our program

1- Test inputting invalid ID in option1:

```
Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
1
please enter the patient ID number : (7-digit number)
115
error in reading or converting number .
please make sure to follow the format !!!

-- program is finished running --
```

2- Test inputting invalid Date in option1:

```
Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
1
please enter the patient ID number : (7-digit number)
1300500
please enter the test type : ( BBB any other formate will not be accepted ! )
BPT
please enter the test date : (YYYY-MM any other formate will not be accepted ! )
22
error in reading or converting number .
please make sure to follow the format !!!

-- program is finished running --
```


3- Test inputting invalid Name in option1:

```
1
please enter the patient ID number : (7-digit number)
1234567
please enter the test type : ( BBB any other formate will not be accepted ! )
1f1
please enter the test date : (YYYY-MM any other formate will not be accepted ! )
2020-12
please enter the test result : (x.x any other format will not be accepted)
123.1
error in test name !!
```

4- Test inputting invalid ID in option2:

```
Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
2
please enter the patient ID number : (7-digit number)
11
error in reading or converting number .
please make sure to follow the format !!!

-- program is finished running --
```

5- Test inputting invalid TestName in option3:

```
Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
3
Enter the name of the test: FFF
The name you entered is incorrect, Please check your inforamtion and try again :)
```

6- Test inputting invalid ID & TestName & Date in option5:

```
Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
5
Enter the ID:11
TestName:BPT
Date:2021-06
Please enter the first NewResult: 55

Please enter the second NewResult: 66
Failed to update the test, check the data you entered and try again
```

```
Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
5
Enter the ID:1300500
TestName:GGG
Date:2021-09
Enter a valid test name
```

```
Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
5
Enter the ID:1300500
TestName:HGB
Date:20

Please enter the NewResult: 55
Failed to update the test, check the data you entered and try again
```

7- Test inputting invalid ID & TestName & Date in option6:

```
Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
6
ID:11
TestName:HGB
Date:2021-08
The informaion you have entered was not found, please check your data and try again :)

Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
6
ID:1300500
TestName:iii
Date:2021-03
Invalid Test name

Please choose an option:
1) Add a new medical test
2) Search for a patient by ID
3) Search for upnormal tests
4) Find average test values
5) Update an existing test result
6) Delete a test
7) Exit
6
ID:1300500
TestName:HGB
Date:20
The informaion you have entered was not found, please check your data and try again :)
```